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LITHOGRAPHY SYSTEM AND METHODS

Abstract

A method includes: depositing a mask layer over a substrate; protecting a mask of a mask assembly by a pellicle attached to a pellicle assembly, the pellicle and the pellicle assembly being laterally offset from the mask assembly by a distance; directing first radiation toward the mask with the pellicle positioned overlapping the mask; and exposing the mask layer of a semiconductor wafer by second radiation carrying a pattern of the mask

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Background/Summary

BACKGROUND

[0001] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIGS. **1**A and **1**B are views of portions of a lithography scanner according to embodiments of the present disclosure.

[0004] FIGS. **2**A-**2**C are views illustrating effects of particles on a mask according to various aspects of the present disclosure.

[0005] FIGS. **3**A and **3**B are views illustrating use of a pellicle in accordance with various embodiments.

[0006] FIGS. **4**A-**4**C are views illustrating a system including a pellicle and mask in accordance with various embodiments.

[0007] FIGS. **5** and **6** are views illustrating methods of fabricating a device according to various aspects of the present disclosure.

DETAILED DESCRIPTION

[0008] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0009] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0010] Terms such as “about,” “roughly,” “substantially,” and the like may be used herein for ease of description. A person having ordinary skill in the art will be able to understand and derive meanings for such terms.

[0011] The present disclosure is generally related to lithography equipment for fabricating semiconductor devices, and more particularly to a pellicle-less frame that is part of a mask assembly. Dimension scaling (down) is increasingly difficult in advanced technology nodes. Lithography techniques employ ever shorter exposure wavelengths, including deep ultraviolet (DUV; about 193-248 nanometers), extreme ultraviolet (EUV; about 10-100 nanometers; particularly 13.5 nanometers), and X-ray (about 0.01-10 nanometers) to ensure accurate patterning at the scaled-down dimensions. In an EUV scanner, EUV light is generated by a light source, and reflected toward a wafer by multiple mirrors and a reflective mask. Only a fraction of the EUV light reaches the wafer, such that increasing intensity of EUV light generated by the light source is a topic of much interest.

[0012] The EUV scanner includes a mask assembly for shaping and reflecting light from a light source that is incident on the mask assembly. Elimination of particle sources inside a scanner chamber is beneficial to improving yield by avoid defects that arise from particles attaching to a mask of the mask assembly. However, even with significant progress on particle reduction, a pellicle that protects the mask from particles continues to be used in manufacture of some devices.

[0013] When a pellicle is employed, the pellicle is directly mounted as a film in front of the mask. In previous technology generations, ArF- and KrF-type pellicles could be manufactured to thickness of about 1 um, which provided strong reliability, because light transmission through material of the pellicle could reach about 99%. However, EUV light used in current and next generation technologies is very easily absorbed by any material. As such, the film thickness for a pellicle used in an EUV scanner is about 80 nm, and reliability is a concern as ruptures are common under very fast motion during scanning exposure. The pellicle is directly mounted in front of the mask. However, the very thin pellicle film mounted on the mask sustains scanning motion that may exceed 20G during exposure. Such a heavy acceleration speed is very challenging for pellicle reliability. Increasing thickness of the pellicle is one possibility, but negatively impacts EUV productivity.

[0014] In embodiments of this disclosure, a novel mechanical assembly is included in a scanner to achieve a “stand-alone” pellicle. The pellicle is not directly mounted on the EUV mask but can still be used to protect the mask from particles. In the embodiments, the pellicle film is separated from the mask to have non-synchronized movement, so as to escape the heavy acceleration speed during scanner exposure process, which increases sustainability during usage. In some embodiments, the pellicle is mounted on the mask but not completely fixed, thereby providing some tolerance during high-speed motion for more sustainability. The film size of the pellicle may be 1 cm×10 cm to 30 cm×30 cm. The pellicle having a larger size of film and the non-synchronized movement mechanical device may be used for shifting, rotating or performing other motion changes to expand applications, such as if the film suffers damage from particle sources or for enhancing protection of selected areas of the mask.

[0015] FIG. **1**A is a diagrammatic schematic view of a lithography exposure system **10**, in accordance with some embodiments. Description of the lithography system **10** is given in broad terms to provide context for understanding the embodiments of the present disclosure. In some embodiments, the lithography exposure system **10** is an extreme ultraviolet (EUV) lithography system designed to expose a resist layer by EUV radiation and may also be referred to as the EUV system **10**. The EUV system **10** may also be referred to as an EUV scanner or lithography scanner. The lithography exposure system **10** includes a light source **120**, an illuminator **140**, a mask stage **16**, a projection optics module (or projection optics box (POB)) **130** and a substrate stage **24**, in accordance with some embodiments. The elements of the lithography exposure system **10** can be added to or omitted, and the disclosure should not be limited by the embodiment.

[0016] The light source **120** is configured to generate light radiation having a wavelength ranging between about 1 nm and about 100 nm in certain embodiments. In one particular example, the light source **120** generates an EUV radiation with a wavelength centered at about 13.5 nm. Accordingly, the light source **120** is also referred to as an EUV radiation source. However, it should be appreciated that the light source **120** should not be limited to emitting EUV radiation. The light source **120** can be utilized to perform any high-intensity photon emission from excited target fuel.

[0017] In various embodiments, the illuminator **140** includes various refractive optic components, such as a single lens or a lens system having multiple reflectors **100**, for example lenses (zone plates) or alternatively reflective optics (for EUV lithography exposure system), such as a single mirror or a mirror system having multiple mirrors in order to direct light **86** from the light source **120** onto the mask stage **16**, particularly to a mask **18** secured on the mask stage **16**. In embodiments in which the light source **120** generates light in the EUV wavelength range, reflective optics and/or lenses are employed. In some embodiments, the illuminator **140** includes at least two lenses, at least three lenses, or more.

[0018] The mask stage **16** is configured to secure the mask **18**. In some embodiments, the mask stage **16** includes an electrostatic chuck (e-chuck) to secure the mask **18**. One reason an e-chuck is beneficial is that gas molecules absorb EUV radiation and the e-chuck is operable in the lithography exposure system for the EUV lithography patterning that is maintained in a vacuum environment to avoid EUV intensity loss. In the present disclosure, the terms mask, photomask, and reticle are used interchangeably. In the present embodiment, the mask **18** is a reflective mask. One exemplary structure of the mask **18** includes a substrate with a suitable material, such as a low thermal expansion material (LTEM) or fused quartz. In various examples, the LTEM includes TiO.sub.2 doped SiO.sub.2, or other suitable materials with low thermal expansion. The mask **18** includes a reflective multilayer deposited on the substrate. The mask stage **16** is operable to translate in two horizontal directions, such as an X-axis direction and a Y-axis direction, so as to expose multiple different regions of the semiconductor wafer **22** to light having a pattern generated by the mask **18**. The semiconductor wafer **22** may have a mask layer **26** thereon, which may be a photoresist layer that is sensitive to the light carrying the pattern of the mask **18**.

[0019] The projection optics module (or projection optics box (POB)) **130** is configured for imaging the pattern of the mask **18** on to a semiconductor wafer **22** secured on the substrate stage **24** of the lithography exposure system **10**. In some embodiments, the POB **130** has refractive optics (such as for a UV lithography exposure system) or alternatively reflective optics (such as for an EUV lithography exposure system) in various embodiments. The light directed from the mask **18**, carrying the image of the pattern defined on the mask, is collected by the POB **130**. The illuminator **140** and the POB **130** are collectively referred to as an optical module of the lithography exposure system **10**. In some embodiments, the POB **130** includes at least six reflective optics.

[0020] In some embodiments, the semiconductor wafer **22** may be made of silicon or other semiconductor materials. Alternatively or additionally, the semiconductor wafer **22** may include other elementary semiconductor materials such as germanium (Ge). In some embodiments, the semiconductor wafer **22** is made of a compound semiconductor such as silicon carbide (SiC), gallium arsenic (GaAs), indium arsenide (InAs), or indium phosphide (InP). In some embodiments, the semiconductor wafer **22** is made of an alloy semiconductor such as silicon germanium (SiGe), silicon germanium carbide (SiGeC), gallium arsenic phosphide (GaAsP), or gallium indium phosphide (GaInP). In some other embodiments, the semiconductor wafer **22** may be a silicon-on-insulator (SOI) or a germanium-on-insulator (GOI) substrate.

[0021] In addition, the semiconductor wafer **22** may have various device elements. Examples of device elements that are formed in the semiconductor wafer **22** include transistors (e.g., metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high voltage transistors, high-frequency transistors, p-channel and/or n-channel field-effect transistors (PFETs/NFETs), nanostructure field-effect transistors, and the like), diodes, and/or other applicable elements. Various processes are performed to form the device elements, such as deposition, etching, implantation, photolithography, annealing and/or other suitable processes. In some embodiments, the semiconductor wafer **22** is coated with a resist layer sensitive to the EUV radiation. Various components including those described above are integrated together and are operable to perform lithography processes.

[0022] The lithography exposure system **10** may further include other modules or be integrated with (or be coupled with) other modules, such as a cleaning module designed to provide hydrogen gas to the light source **120**. The hydrogen gas helps reduce contamination in the light source **120**. Further description of the light source **120** is provided with reference to FIG. **1**B.

[0023] In FIG. **1**B, the light source **120** is shown in a diagrammatical view, in accordance with some embodiments. In some embodiments, the light source **120** employs a dual-pulse laser produced plasma (LPP) mechanism to generate plasma **88** and further generate EUV radiation from the plasma. The light source **120** includes a droplet generator **30**, a droplet receptacle **35**, a laser generator **50**, a laser produced plasma (LPP) collector **60**, a monitoring device **70** and a controller **90**. Some or all of the above-mentioned elements of the light source **120** may be held under vacuum. It should be appreciated that the elements of the light source **120** can be added to or omitted and should not be limited by the embodiment.

[0024] The droplet generator **30** is configured to generate a plurality of droplets **82**, which may be elongated, of a target fuel **80** to a zone of excitation at which at least one laser pulse **51** from the laser generator **50** hits the droplets **82** along an X-axis, as shown in FIG. **1**B. In an embodiment, the target fuel **80** includes tin (Sn). In an embodiment, the droplets **82** may be formed with an elliptical shape. In an embodiment, the droplets **82** are generated at a rate of about 50 kilohertz (kHz) and are introduced into the zone of excitation in the light source **120** at a speed of about 70 meters per second (m/s). Other material can also be used for the target fuel **80**, for example, a tin containing liquid material such as eutectic alloy containing tin, lithium (Li), and xenon (Xe). The target fuel **80** in the droplet generator **30** may be in a liquid phase.

[0025] The laser generator **50** is configured to generate at least one laser pulse to allow the conversion of the droplets **82** into plasma **88**. In some embodiments, the laser generator **50** is configured to produce a laser pulse **51** to the lighting point **52** to convert the droplets **82** to plasma **88** which generates EUV radiation **84**. The laser pulse **51** is directed through window (or lens) **55** and irradiates droplets **82** at the lighting point **52**. The window **55** is formed in the collector **60** and adopts a suitable material substantially transparent to the laser pulse **51**. The droplet receptacle **35** catches and collects unused droplets **82** and/or scattered material of the droplets **82** resulting from the laser pulse **51** striking the droplets **82**.

[0026] The plasma emits EUV radiation **84**, which is collected by the collector **60**. The collector **60** further reflects and focuses the EUV radiation **84** for the lithography processes performed through an exposure tool. In some embodiments, the collector **60** has an optical axis **61** which is parallel to the Z-axis and perpendicular to the X-axis. The collector **60** may include a single section, as shown, or at least two sections that are offset from each other in the z-axis direction. The collector **60** may further include a vessel wall **65** having first and second pumps **66**, **68** attached thereto. In some embodiments, the first and second pumps **66**, **68** include scrubbers configured to remove particulates and/or gases from the collector **60**. The first and second pumps **66**, **68** may be collectively referred to as "the pumps **66**, **68**" herein.

[0027] In an embodiment, the laser generator **50** is a carbon dioxide (CO2) laser source. In some embodiments, the laser generator **50** is used to generate the laser pulse **51** with single wavelength. The laser pulse **51** is transmitted through an optic assembly for focusing and determining incident angle of the laser pulse **51**. In some embodiments, the laser pulse **51** has a spot size of about 200-300 μm, such as 225 μm. The laser pulse **51** is generated to have certain driving power to meet wafer production targets, such as a throughput of 125 wafers per hour (WPH). For example, the laser pulse **51** is equipped with about 23 KW driving power. In various embodiments, the driving power of the laser pulse **51** is at least 20 kW, such as 27 kW.

[0028] The monitoring device **70** is configured to monitor one or more conditions in the light source **120** so as to produce data for controlling configurable parameters of the light source **120**. In some embodiments, the monitoring device **70** includes a metrology tool **71** and an analyzer **73**. In cases where the metrology tool **71** is configured to monitor condition of the droplets **82** supplied by the droplet generator **30**, the metrology tool may include an image sensor, such as a charge coupled device (CCD), complementary metal oxide semiconductor sensor (CMOS) sensor or the like. The metrology tool **71** produces a monitoring image including image or video of the droplets **82** and transmits the monitoring image to the analyzer **73**. In cases where the metrology tool **71** is configured to detect energy or intensity of the EUV light **84** produced by the droplet **82** in the light source **12**, the meteorology tool **71** may include a number of energy sensors. The energy sensors may be any suitable sensors that are able to observe and measure energy of electromagnetic radiation in the ultraviolet region.

[0029] The analyzer **73** is configured to analyze signals produced by the metrology tool **71** and outputs a detection signal to the controller **90** according to an analyzing result. For example, the analyzer **73** includes an image analyzer. The analyzer **73** receives the data associated with the images transmitted from the metrology tool **71** and performs an image analysis process on the images of the droplets **82** in the excitation zone. Afterwards, the analyzer **73** sends data related to the analysis to the controller **90**. The analysis may include a flow path error or a position error.

[0030] In some embodiments, two or more metrology tools **71** are used to monitor different conditions of the light source **120**. One is configured to monitor condition of the droplets **82** supplied by the droplet generator **30**, and the other is configured to detect energy or intensity of the EUV light **84** produced by the droplet **82** in the light source **120**. In some embodiments, the metrology tool **71** is a final focus module (FFM) and positioned in the laser source **50** to detect light reflected from the droplet **82**.

[0031] The controller **90** is configured to control one or more elements of the light source **120**. In some embodiments, the controller **90** is configured to drive the droplet generator **30** to generate the droplets **82**. In addition, the controller **90** is configured to drive the laser generator **50** to fire the laser pulse **51**. The generation of the laser pulse **51** may be controlled to be associated with the generation of droplets **82** by the controller **90** so as to make the laser pulse **51** hit each target **82** in sequence.

[0032] In some embodiments, the droplet generator **30** includes a reservoir **31** and a nozzle assembly **32**. The reservoir **31** is configured for holding the target material **80**. In some embodiments, one gas line is connected to the reservoir **31** for introducing pumping gas, such as argon, from a gas source **40** into the reservoir **31**. By controlling the gas flow in the gas line, the pressure in the reservoir **31** can be manipulated. For example, when gas is continuously supplied into the reservoir **31** via the gas line, the pressure in the reservoir **31** increases. As a result, the target material **80** in the reservoir **31** can be forced out of the reservoir **31** in the form of droplets **82**.

[0033] FIGS. **2**A-**2**C are views depicting a mask assembly **200** of a lithography scanner, such as the lithography system **10**, and particles **250** according to various aspects of the present disclosure. FIG. **2**A is a side view of a mask assembly **200**. FIG. **2**B is a top view of a mask pattern **230** of the mask assembly **200**. FIG. **2**C is a diagram illustrating exposure errors in regions **225** of a semiconductor wafer **220** due to particles **250**.

[0034] In FIG. **2**A, the mask assembly **200** includes a mask stage **216** and a mask **218** attached thereto. The mask stage **216** and the mask **218** may be the mask stage **16** and the mask **18**, respectively, of FIGS. **1**A and **1**B. The mask **218** includes mask patterns **230** that may be located in a layer of the mask **218** facing reflectors of the illuminator **140** and the POB **130** on either side of the mask assembly **200**.

[0035] Particles **250** may be present in the lithography scanner. The particles **250** may include different types of particles generated by different sources in the lithography scanner. For example, the particles **250** may include tin particles generated by the light source **120** during formation of the plasma **88**. The particles **250** may include SiC particles generated by movement of the mask assembly **200** in the X- and Y-axis directions. The particles **250** may include carbon particles generated by a pod or carrier used for transporting the mask **218** in and out of the lithography scanner. Other particles **250** having different material composition may be generated by other sources internal or external to the lithography scanner. One or more of the particles **250** may settle on the surface of the mask **218** on one or more mask pattern regions of the mask patterns **230**.

[0036] FIG. **2**B shows a view of the mask patterns **230** with a particle **250** thereon. The mask patterns **230** are exposed to the internal environment of the lithography scanner. While the mask assembly **200** is in the lithography scanner, the particle **250** may fall or settle on the mask **218**. It should be understood that "fall" and "settle" include the meaning that a particle drifts toward the mask **218**, makes contact therewith and attaches thereto, regardless of orientation of the mask **218** in three-dimensional space relative to the direction of gravity. The particle **250** may form a short circuit or bridge or merger between one or more pattern regions of the mask patterns **230**. When the pattern of the mask **218** is transferred to a semiconductor wafer, an electrical defect, such as a short circuit or bridge or merger, may occur between features of the semiconductor wafer. For example, neighboring semiconductor fins or neighboring conductive traces may merge unintentionally, which may result in failure of an integrated circuit die formed in the semiconductor wafer.

[0037] FIG. **2**C is a diagrammatic view of a semiconductor wafer **220**, which may be the semiconductor wafer **22** of FIGS. **1**A and **1**B. The view of FIG. **2**C may be a diagram of an image generated by a metrology tool that analyzes the semiconductor wafer **220**. During exposure, in which light carrying the pattern of the mask patterns **230** is incident on the semiconductor wafer **220**, the particle **250** alters the pattern, which is transferred repeatedly onto the semiconductor wafer **220**. As such, quality of the semiconductor wafer **220** is reduced, reducing productivity of the lithography scanner.

[0038] To prevent particles **250** from attaching to the mask patterns **230**, a pellicle may be included that blocks the particles **250**. FIGS. **3**A and **3**B depict a pellicle **370** that is attached to the mask assembly **200**, and description thereof is provided as context for understanding the embodiments of the present disclosure. Generally, when the pellicle **370** is attached (e.g., mounted) to the mask assembly **200** as depicted in FIGS. **3**A and **3**B, the pellicle **370** experiences the same high acceleration forces as the mask assembly **200** along the X- and Y-axis directions due to rapid motion of the mask assembly **200**. Namely, a first rate of acceleration of the mask assembly **200** (or the mask **218**) is the same as a second rate of acceleration of the pellicle **370**. For example, during scanning, the mask assembly **200** may move the mask patterns **230** smoothly across a narrow slit of EUV light while the wafer stage moves the wafer in an opposing direction, so as to expose the wafer to the entirety of the mask patterns **230**. When finished, the mask assembly **200** may rapidly decelerate and/or change direction in the orthogonal direction in the horizontal plane. The mask assembly **200** may experience acceleration at a level of 15G, 20G or more. The pellicle **370** may also experience the same acceleration at a level of 15G, 20G or more. As such, the pellicle **370** may rupture easily due to its thin thickness.

[0039] FIGS. **3**A and **3**B are diagrammatic views showing a pellicle **370** and offset structure **360** installed on the mask assembly **200** to prevent the particles **250** from attaching to the mask **218**. In FIGS. **3**A and **3**B, the pellicle **370** is shown suspended by the offset structure **360** over the mask patterns **230**. The pellicle **370** may be a nanoscale thickness thin film that has high transparency (e.g., >90%) to EUV wavelengths (e.g., 13.5 nm). For example, the pellicle **370** may have thickness in the Z-axis direction in a range of about 50 nm to about 200 nm.

[0040] The offset structure **360** has height D**1**, which may be in a range of about 1 millimeter to about 3 millimeters, or more. The height D**1** is about the same as a separation distance between the pellicle **370** and the mask patterns **230** in the Z-axis direction. The offset structure **360** may have rectangular (e.g., square) shape in the XY-plane, as shown in FIG. **3**B. The offset structure **360** may be adjacent to the mask patterns **230** on four sides, as shown. The offset structure **360** may be offset horizontally in the X-axis and Y-axis directions from the mask patterns **230**. For example, the offset structure **360** may be offset from the mask patterns by a second distance D**2** in the Y-axis direction and by a third distance D**3** in the X-axis direction. The distances D**1**, D**2**, D**3** may be the same as each other. In some embodiments, one or more of the distances D**1**-D**3** is different from others of the distances D**1**-D**3**. For example, the distance D**1** may be in the range of about 1-3 millimeters as described above, and the second and third distances D**2**, D**3** may be in a range of about 0.5 millimeters to about 10 millimeters. Mounting the pellicle **370** on the offset structure **360** can prevent mask defects formed by particles released by the lithography scanner. Generally, the distance D**1** is sufficiently large such that any particle **250** less than a selected size (e.g., diameter less than about 500 nm) that settles on the outside surface of the pellicle **370** is far enough from a focus plane of incident light that the particle **250** does not cause a pattern defect.

[0041] The pellicle **370** may be operated for a selected number of wafers before being replaced. For example, the pellicle **370** may be said to have a "lifetime" of about 10,000 wafers, about 15,000 wafers, or the like. During manufacture of integrated circuit dies on the semiconductor wafer **220**, the mask assembly **200** may translate back and forth along the XY plane, and particles may attach to the outside surface of the pellicle **370**. The particles may include tool particles and pod particles, among other particle types described above with reference to FIGS. **2**A-**2**C. Over time, as the particles accumulate on the pellicle **370**, and due to repeated acceleration along the XY plane of the pellicle **370**, the pellicle **370** may deform. Eventually, the pellicle **370** may rupture and is replaced.

[0042] When the pellicle **370** is mounted to the mask assembly **200** as depicted in FIGS. **3**A and **3**B, the pellicle **370** is subject to high acceleration forces due to rapid motion of the mask assembly **200** along the X- and Y-axis directions as described previously. As such, the pellicle **370** may rupture easily due to its thin thickness.

[0043] FIGS. **4**A-**4**C are diagrams illustrating a partial structure of a lithography system **400** in accordance with various embodiments. In the embodiments, motion of a pellicle **490** is decoupled either completely or partially from motion of a mask stage **416**, which is beneficial to reduce acceleration forces experienced by the pellicle **490** and thereby extend lifespan of the pellicle **490**.

[0044] Another benefit is that the pellicle **490** may be formed with a thinner thickness, which may improve wafer throughput of a scanner that includes the pellicle **490** due to less power loss through the pellicle **490**.

[0045] Decoupling the pellicle **490** from the mask stage **416** provides a further benefit that rework time may be decreased. This is because, instead of having to perform a complex process to remove (e.g., demount and remount) the pellicle **490** from the mask stage **416**, the pellicle **490** may be easily discarded and replaced with a new pellicle simply by motion of, for example, a robot arm.

[0046] Another benefit is that acceleration of the mask stage **416** may be increased due to the acceleration not being applied to the pellicle **490**.

[0047] Glue or adhesive selection for mounting the pellicle may also be more varied, which may eliminate use of specialized adhesives that are costly, hard to produce or have limited supply.

[0048] Another benefit is that instead of using one pellicle per mask in a semiconductor fab, one pellicle may be provided for each tool, which is beneficial to efficient pellicle usage and stocking. For example, number of pellicles used may decrease from about 200 pieces to about 6 pieces.

[0049] A pellicle swapping system including an atmosphere-vacuum interface chamber, turbo pump, and two sets of load-lock gate valves for vacuum-side and atmosphere-side is also provided. This is beneficial to improve flexibility to select application of a pellicle or not via recipe. For example, a higher power output of the EUV light may be paired with a thicker or higher-quality pellicle. In another example, higher power EUV light may be paired with faster scan speed of the mask assembly during exposure.

[0050] As another benefit, a vacuum system may be provided in communication with a pellicle handler to exhaust debris when facing rupture of the pellicle.

[0051] FIGS. **4**A-**4**C depict a system **400** in accordance with various embodiments. FIG. **4**A is a diagrammatic block diagram of the system **400**. FIG. **4**B is a side view diagram of the system **400**. FIG. **4**C is a further side view diagram of the system **400** illustrating an operation thereof.

[0052] The system **400** may include a first chamber or "swapping chamber" **410**, a first transfer chamber or "first lock and load chamber" or "pellicle lock and load chamber" **420**, a pellicle handler **430**, a second chamber or "reticle clamping chamber" **440**, a storage **450**, a third chamber **460** and a second transfer chamber or "second lock and load chamber" or "mask lock and load chamber."

[0053] The first chamber **410** may be a chamber that is at atmospheric pressure and may store one or more pellicles **490**. A first transfer assembly or "first robot arm" may be positioned in the first chamber **410**. The first robot arm may be operable to pick a pellicle **490** and move the pellicle **490** to an interface with the pellicle lock and load chamber **420**.

[0054] The second chamber **440** may be a chamber that is at vacuum or near-vacuum pressure. A reticle stage or mask stage **416** may be positioned in the second chamber **440**. A second transfer assembly may be positioned in the second chamber **440**. The second transfer assembly may be operable to move the mask stage **416**. For example, the mask stage **416** may be moved by the second transfer assembly during EUV scanning and/or during mounting of a mask **418** thereto. Exposure by EUV light scanning may be performed while the mask stage **416** and optionally the pellicle **490** are positioned in the second chamber **440**.

[0055] The third chamber **460** may be a chamber that is at atmospheric pressure and may store one or more masks **418** (depicted in FIGS. **4**B and **4**C) or be in communication with the storage **450** that stores the one or more masks **418**. In some embodiments, the storage **450** is included in the third chamber **460**. A third transfer assembly or "second robot arm" may be positioned in the third chamber **460**. The second robot arm may be operable to pick a mask **418** and move the mask **418** to an interface with the mask lock and load chamber for mounting the mask **418** to the mask stage **416**.

[0056] The system **400** includes two lock and load chambers and may include at least one pump. The pump is in communication with the pellicle lock and load chamber **420**. The lock and load chambers are beneficial to pressurize the pellicle **490** and the mask **418**, respectively, during transfer between the second chamber **440** and the first chamber **410** and between the second chamber **440** and the third chamber **460**, respectively. A fourth transfer assembly may be positioned in the pellicle lock and load chamber **420**. The fourth transfer assembly is operable to pick the pellicle **490** from the first robot arm and to place the pellicle **490** on the pellicle handler **430**. A fifth transfer assembly may be positioned in the mask lock and load chamber. The fifth transfer assembly is operable to pick the mask **418** from the second robot arm and transfer the mask **418** to the mask stage **416**.

[0057] The pellicle **490** may be stored in the first chamber **410** at atmospheric pressure and the second chamber **440** may be at vacuum or near-vacuum pressure. When transferring the pellicle **490** from the first chamber **410** to the second chamber **440**, the pellicle lock and load chamber **420** may initially be at atmospheric pressure or another pressure that matches pressure inside the first chamber **410**. Once the pellicle **490** is positioned inside the pellicle lock and load chamber **420**, the pump may pull down pressure in the pellicle lock and load chamber **420** to a pressure that matches pressure inside the second chamber **440** (e.g., vacuum or near-vacuum). The pellicle **490** may then be transferred to the second chamber **440** from the pellicle lock and load chamber **420**.

[0058] The mask **418** may be stored in the third chamber **460** at atmospheric pressure and the second chamber **440** may be at vacuum or near-vacuum pressure. When transferring the mask **418** from the third chamber **460** to the second chamber **440**, the mask lock and load chamber may initially be at atmospheric pressure or another pressure that matches pressure inside the third chamber **460**. Once the mask **418** is positioned inside the mask lock and load chamber, a pump (not shown) may pull down pressure in the mask lock and load chamber to a pressure that matches pressure inside the second chamber **420** (e.g., vacuum or near-vacuum). The mask **418** may then be transferred to the second chamber **440** from the mask lock and load chamber.

[0059] FIG. **4**B is a side view diagram depicting the mask assembly **41** and the pellicle assembly **49** in accordance with various embodiments. The pellicle assembly **49** may be an embodiment of the pellicle handler **430**, and holds the pellicle **490** and is operable to translate the pellicle **490** in at least one horizontal direction (e.g., an X-axis direction depicted in FIG. **4**D), independent of the mask assembly **41**. The mask assembly **41** includes the mask stage **416** and the mask **418**, and is operable to translate in the first direction independent of the pellicle **490**. In some embodiments, the mask assembly **41** is operable to translate in two directions, such as the first direction (e.g., the X-axis direction of FIG. **4**D) and a second direction (e.g., Y-axis direction of FIG. **4**D) that is transverse (e.g., orthogonal to) the first direction. The second direction may be a horizontal direction, whereas the pellicle assembly **49** may be offset from the mask assembly **41** along a vertical direction, such as a Z-axis direction depicted in FIG. **4**D. The mask stage **416** may be similar to the mask stage **16** described with reference to FIG. **1**A. The mask **418** may be similar to the mask **18** described with reference to FIG. **1**A. The pellicle **490** may be similar to the pellicle **370** described with reference to FIG. **3**A.

[0060] The pellicle assembly **49** includes an actuator, which may be a linear actuator. The actuator includes a housing **492** and one or more rods **494** that are operable to move in and out of the housing **492** to provide motion along a single axis. In some embodiments, the actuator is a multidirectional actuator that provides motion along two or more axes, such as an XY linear stage actuator, gantry actuator or the like. The pellicle **490** is connected to the rods **494** and may be positioned based on extension and contraction of the rods **494**. Control of extension and contraction of the rods **494** may be mechanical/electro mechanical, hydraulic, pneumatic, piezoelectric or the like. While not depicted, one or more of a screw (e.g., a ballscrew), spring, belt, motor or the like may be positioned in the housing **492** to provide control of extension and contraction of the rods **494**.

[0061] The pellicle **490** may be offset from the mask **418** in the vertical Z-axis direction by a distance that is in a range of about 1 mm to about 10 mm, such as about 1 mm to about 3 mm. The pellicle **490** may have dimensions (e.g., width and length) in the first and second directions of about 1 cm×10 cm to about 30 cm×30 cm. In some embodiments, the mask **418** has a field size that is less than about 26 mm×33 mm or less than about 26 mm×16.5 mm. The pellicle **490** may have dimensions the same as those of the field size or larger. For example, the mask **418** may have a length in the X-axis direction of 33 mm, such that the mask assembly **41** may traverse a distance of about 66 mm for each scan of the mask **418**. The pellicle **490** may have length the same as the distance over which the mask **418** traverses (e.g., about 66 mm) or longer. In some embodiments, the length of the pellicle **490** is between the length of the field size of the mask **418** and about twice the length of the field size of the mask **418**. The length of the pellicle **490** being about the same as the field size may mean that the pellicle **490** shifts at about the same rate or acceleration as the mask **418** during scanning. The length of the pellicle **490** exceeding twice that of the field size may mean that the pellicle **490** shifts very little during scanning, but other difficulties such as warpage of the pellicle **490** may be more pronounced with larger area of the pellicle **490**. Width of the pellicle **490** may be about the same as width of the field size of the mask **418** or even slightly larger to allow for reduced shifting of the pellicle **490** in the Y-axis direction.

[0062] In accordance with various embodiments, the pellicle **490** may be attached or mounted to a frame, which may be attached or mounted to an extension plate of the pellicle assembly **49**. The extension plate may be attached to a rod plate, which is attached to the rods **494**. The extension plate is offset from the rod plate along the first direction. The frame may attach to the extension plate by a fastener (e.g., one or more screws), one or more magnets, by sliding into a slot of the extension plate or in another appropriate manner. The frame may hold the pellicle **490** and may optionally stretch the pellicle **490** such that the pellicle **490** is taut.

[0063] FIG. **4**C is another side view of the pellicle assembly **49** and the mask assembly **41** during scanning by light **860**, in accordance with various embodiments. The light **860** may be similar to the light **86** described with reference to FIG. **1**A. The light **860** may impinge on the mask **418** through the pellicle **490** held by the pellicle assembly **49**. Although the light **860** is depicted as illuminating the entire mask **418**, the light **860** may instead be a slit-shaped beam that illuminates a thin line of the mask **418**, for example, along the Y-axis direction. The mask assembly **41** may move back and forth along the X-axis direction to reflect the light **860** from the entire surface of the mask **418**. The pellicle **490** protects the mask **418** from particles, and may move independently of the mask assembly **41** due to not being mounted to the mask assembly **41**.

[0064] FIG. **5** is a flowchart of a process **501** for forming a device in accordance with various embodiments. In some embodiments, the process **501** for forming the device includes a number of operations (**500**, **510**, **520**, **530**, **540**, **550**, **560**, **570** and **580**). FIG. **6** is a flowchart of a process **601** for replacing a pellicle in accordance with various embodiments. In some embodiments, the process **601** includes a number of operations (**600**, **610**, **620**, **630**, **640**, **650**, **660**, **670**, **680** and **690**). The processes **501**, **601** will be further described according to one or more embodiments. It should be noted that the operations of the processes **501**, **601** may be rearranged or otherwise modified from what is depicted in FIGS. **5** and **6** within the scope of the various aspects. It should further be noted that additional processes may be provided before, during, and after the processes **501**, **601** and that some other processes may be only briefly described herein. In some embodiments, the processes **501**, **601** are performed by the lithography exposure system **10** described in FIGS. **1**A-**4**C. The embodiments are described with reference to the structural elements described in FIGS. **1**A-**4**C, but the processes **501**, **601** may be performed by a lithography system having one or more structural elements that are different from those of the lithography system **10**. Some of the operations of the processes **501**, **601** may be performed or controlled by the controller **90** described with reference to FIG. **1**A.

[0065] In operation **500**, a mask layer (e.g., a mask layer **26** shown in FIG. **1**A) is deposited over a substrate. In some embodiments, the mask layer **26** includes a photoresist layer that is sensitive to the EUV radiation reflected by the mask **418**. In some embodiments, the substrate is a semiconductor substrate, such as the semiconductor wafer **22** described with reference to FIGS. **1**A and **1**B. In some embodiments, the substrate is a layer overlying the semiconductor substrate, such as a dielectric layer, a metal layer, a hard mask layer, or other suitable layer. In some embodiments, the mask layer is deposited by spin coating or other suitable process.

[0066] In operation **510**, a pellicle (e.g., the pellicle **490**) is arranged offset from a mask (e.g., the mask **418**). The pellicle **490** may be offset from the mask **418** in the Z-axis direction, as depicted in FIG. **4**B. The pellicle **490** may overlap the mask **418** in the XY plane, for example. In operation **510**, the pellicle **490** is not attached to the mask assembly **41** and is operable to move independent of motion of the mask assembly **41**. This is described with reference to FIGS. **4**A-**4**C above. Operation **510** may follow operation **500**, in some embodiments. In other embodiments, operation **510** may be performed prior to or simultaneously with operation **500**.

[0067] In operation **520**, following arranging of the pellicle **490**, the mask assembly **41** is translated so as to expose a region of the mask layer **26** on the substrate. For example, the mask assembly **41** may be translated in the X-axis direction, the Y-axis direction, or both. During translation of the mask assembly **400**, the mask **418** thereof is protected by the pellicle **490**. The mask **418** may be protected from particles by the pellicle **490**. The protection may include deflecting one or more of the particles, adhering one or more of the particles, or both. The adhering may be physical by the pellicle **490** itself.

[0068] In optional operation **530**, the pellicle **490** is shifted. For example, the pellicle **490** may be shifted along the X-axis direction, as depicted in FIG. **4**D. In some embodiments, when the pellicle **490** is sufficiently large to cover the entirety of a translation path of the mask **418**, the pellicle **490** may be stationary through the shifting of the mask **418** during scanning. When the pellicle **490** is shifted, the pellicle **490** may be shifted at a lower acceleration or a lower force than the mask **418**. This is because the pellicle **490** is decoupled from the mask **418**. In some embodiments, the pellicle **490** does not shift while the mask **418** shifts, such that the pellicle **490** experiences no force along the XY plane, e.g., the plane of motion of the mask **418**. Shifting the pellicle **490** with reduced or no acceleration or force in the XY plane is beneficial for the reasons described above with reference to FIGS. **4**A-**4**C.

[0069] In operation **540**, radiation is reflected from the collector **60** and directed toward the mask layer **26**. The radiation is reflected along an optical path between the collector **60** and the mask layer **26**, which may be on the semiconductor wafer **22**, such as that illustrated in FIG. **1**A. In some embodiments, the radiation is reflected according to a pattern, such as exists on the mask **418**, which may be a reflective mask. The radiation may be EUV light having wavelength centered at about 13.5 nm. The radiation may pass through the pellicle **490** before reaching the mask **418**. In some embodiments, operation **540** is performed continuously while operation **520** and optional operation **530** are repeated throughout the scan.

[0070] In operation **550**, in response to scanning being completed, e.g., if the entire pattern of the mask **418** has been transferred to the mask layer **26**, the process **501** proceeds to operation **552**. If less than then entire pattern of the mask **418** has been scanned, such that the scan is not complete, the process **501** proceeds to operation **520** to translate the mask **418**.

[0071] In operation **552**, a determination is made whether the entire wafer has been exposed. In response to the wafer being incomplete (e.g., one or more dies or die regions are still unexposed), the wafer stage may translate the wafer to a subsequent die or die region, and the process **501** may proceed to operation **520**.

[0072] In operation **552**, in response to the wafer being complete (e.g., all dies or die regions have been exposed), the process **501** may proceed to operation **560**.

[0073] In operation **560**, openings are formed in the mask layer **26** by removing pattern regions of the mask layer **26** exposed to the radiation. In some embodiments, the openings are formed by removing regions of the mask layer **26** not exposed to the radiation.

[0074] In operation **570**, material of one or more layers underlying the mask layer **26** is removed, forming second openings. The material removed is in regions of the layer exposed by the openings in the mask layer **26**. In some embodiments, the layer is a dielectric layer, a semiconductor layer, or other layer.

[0075] In operation **580**, features are formed in the second openings of the layer. For example, source/drain regions may be epitaxially grown in the second openings. For example, metal traces may be deposited in the second openings. For example, gate structures including a high-k dielectric layer and a metal layer may be formed in the second openings.

[0076] FIG. **6** depicts a flowchart of a process **601** for replacing a pellicle (e.g., the pellicle **490**) in accordance with various embodiments. In some embodiments, the pellicle **490** may be replaced under a number of different conditions. For example, the pellicle **490** may be replaced based on a lifetime of the pellicle **490** expiring. The pellicle **490** may be replaced based on the pellicle **490** being damaged or ruptured. The pellicle **490** may be replaced based on changing from one mask to another different mask. For example, a thinner or thicker pellicle may be used for either of the two masks. The pellicle **490** may be replaced based on a change in recipe. For example, a first exposure power may be used when scanning a first wafer, then a second exposure power may be used when scanning a second wafer. The pellicle **490** may be changed to a thicker or thinner pellicle based on using the first exposure power or the second exposure power. The pellicle **490** may be changed or removed based on a mask not needing protection of a pellicle. The process **601** may be performed by the lithography systems **10**, **400** described with reference to FIGS. **1**A-**4**C, but may also be performed by a similar lithography system that has fewer, more or different components than the lithography systems **10**, **400**.

[0077] In operation **600**, when the pellicle **490** is to be changed or removed, the pellicle handler (e.g., the pellicle assembly **49**) is positioned in a first lock and load chamber (e.g., the pellicle lock and load chamber **420**). In some embodiments, the frame holding the pellicle **490** is picked by the fourth transfer assembly and moved into the pellicle lock and load chamber **420**. In some embodiments, the entire pellicle assembly **49** with the pellicle **490** attached thereto is picked and moved into the pellicle lock and load chamber **420**.

[0078] Operation **610** follows operation **600**. In operation **610**, following positioning the pellicle handler in the first lock and load chamber, a first gate valve on a vacuum side (e.g., a side connected to the second chamber **440**) is closed. At this time, a second gate valve on an atmosphere side (e.g., a side connected to the first chamber **410**) is also closed.

[0079] Operation **620** follows operation **610**. In operation **620**, following closing the first gate valve on the vacuum side, venting by inert gas may be performed. The venting may remove particles that have settled on the pellicle **490** and/or the pellicle handler **49** and may also bring pressure in the first lock and load chamber to a level similar to or the same as that of the first chamber. Operation **620** may be performed by the pump **66**.

[0080] Operation **630** follows operation **620**. In operation **630**, following venting the first lock and load chamber, the second gate valve on the atmosphere side (e.g., that connects the first lock and load chamber **420** to the first chamber **410**) is opened. During operation **630**, the first gate valve remains closed.

[0081] Operation **640** follows operation **630**. In operation **640**, following opening the second gate valve on the atmosphere side, the pellicle handler is positioned in the first chamber. In some embodiments, the frame is moved from the pellicle lock and load chamber **420** to the first chamber **410** by the fourth transfer assembly. The first transfer assembly may pick the frame from the fourth transfer assembly. In some embodiments, the pellicle assembly **49** with the pellicle **490** attached thereto is moved from the first lock and load chamber **420** to the first chamber **410** by the fourth transfer assembly.

[0082] Operation **650** follows operation **640**. In operation **650**, following operation **640**, the first pellicle is replaced with a second pellicle (or optionally with no pellicle). For example, the pellicle **490** (or “first pellicle”) in the frame may be removed from the pellicle assembly **49** and placed in storage or discarded, and another pellicle (or “second pellicle”) in another frame may be picked from storage by the first transfer assembly. Then, the second pellicle may be positioned on the pellicle assembly **49** by the first transfer assembly. The second pellicle may be the same as the pellicle **490**, e.g., the second pellicle may have the same or similar parameters (length, width, thickness, material or the like) as the pellicle **490**. For example, when the pellicle **490** is damaged or expires and is to be replaced, the second pellicle may be substantially the same as the pellicle **490**. In some embodiments, the second pellicle is different from the pellicle **490**. For example, when the lithography system begins a different batch of wafers under a different recipe, the second pellicle may have one or more parameters (e.g., length, width, thickness, material or the like) that are different from those of the pellicle **490**. In some embodiments, the pellicle **490** is replaced with nothing. Namely, the lithography system may begin a batch of wafers for which the recipe does not call for a pellicle. As such, after storing or disposing of the pellicle **490**, one or both of the first and fourth transfer assemblies may operate in a standby mode, waiting for instruction to pick yet another pellicle in a subsequent operation. For example, the pellicle assembly **49** may be positioned in the first chamber **410** while a wafer is being exposed using a mask with no pellicle therebetween.

[0083] Operation **660** follows operation **650**. Following replacement of the first pellicle **490** with the second pellicle, the pellicle handler (e.g., the pellicle assembly **49**) having the second pellicle attached thereto is positioned in the first lock and load chamber (e.g., the first lock and load chamber **420**). In some embodiments, the pellicle assembly **49** is picked by the fourth transfer assembly from the first transfer assembly, and the fourth transfer assembly retracts to position the pellicle assembly **49** inside the first lock and load chamber **420**. At this time, a door or gate valve that connects the first lock and load chamber **420** to the second chamber **440** may be closed.

[0084] Operation **670** follows operation **660**. After the pellicle assembly **49** having the second pellicle attached thereto is positioned in the first lock and load chamber **420**, pressure in the first lock and load chamber **420** is lowered. In some embodiments, the pressure in the first lock and load chamber **420** is lowered to pressure level that is at vacuum or near-vacuum. The pressure in the first lock and load chamber **420** may be lowered to a pressure level that is the same as or about the same as that of the second chamber **440**. When the pressure of the second chamber **440** is at vacuum or near-vacuum, the pressure of the first lock and load chamber **420** may be lowered to vacuum or near-vacuum, respectively in operation **670**. This is beneficial to avoid damage to the second pellicle during transfer of the second pellicle to the second chamber **440**. Pumping down the pressure may also be beneficial to remove particles that may be present on the second pellicle prior to using the second pellicle during exposure. Prior to drawing down the pressure in the first lock and load chamber **420**, a door or gate valve that connects the first lock and load chamber **420** to the first chamber **410** may be closed. Operation **670** may be performed by the pump **66**.

[0085] Operation **680** follows operation **670**. After the pressure of the first lock and load chamber **420** is lowered to at or near that of the second chamber **440**, the gate valve that connects the first lock and load chamber **420** and the second chamber **440** is opened.

[0086] Operation **690** follows operation **680**. After the gate valve is opened, the pellicle handler (e.g., the pellicle assembly **49** with the second pellicle attached thereto) may be positioned in an exposure position. The exposure position may be in the second chamber **440**. The pellicle assembly **49** may be attached to a housing of the second chamber **440**, or may be held in place by another assembly in the second chamber **440** that is attached to the housing. The exposure position may be a position selected such that the second pellicle overlaps a mask that the second pellicle is to protect during exposure of a wafer.

[0087] With the second pellicle in the exposure position, an exposure operation may be performed to transfer a pattern of a mask protected by the second pellicle to a layer (e.g., a resist layer) of a semiconductor wafer.

[0088] Embodiments may provide advantages. The pellicle **490** being decoupled from the mask **418** improves reliability and lifetime of the pellicle **490** due to the pellicle **490** experiencing lower or no force or acceleration compared to the mask **418**. The pellicle **490** being held by the pellicle handler or pellicle assembly **49** allows for a single pellicle **490** to be used with multiple masks **418**, improving efficiency of use of pellicles **490** in the lithography exposure system **10**. The pellicle assembly **49** may reduce rework and allow for rapid discarding and replacement when the pellicle **490** is at its end of life, is damaged or is ruptured.

[0089] In accordance with at least one embodiment, a method includes: depositing a mask layer over a substrate; protecting a mask of a mask assembly by a pellicle attached to a pellicle assembly, the pellicle and the pellicle assembly being laterally offset from the mask assembly by a distance; directing first radiation toward the mask with the pellicle positioned overlapping the mask; and exposing the mask layer of a semiconductor wafer by second radiation carrying a pattern of the mask.

[0090] In accordance with at least one embodiment, a method includes: exposing a first semiconductor wafer by performing a first exposure operation using a first mask and a first pellicle, the first pellicle being attached to a pellicle assembly, the first pellicle and the pellicle assembly being laterally offset from the first mask by a distance, the pellicle assembly including an actuator; removing the first pellicle from the pellicle assembly; after the removing, attaching a second pellicle to the pellicle assembly, the second pellicle being a different pellicle than the first pellicle; and exposing a second semiconductor wafer by performing a second exposure operation using the second pellicle.

[0091] In accordance with at least one embodiment, a method includes: a light source that, in operation, generates light; a mask having a pattern, the mask, in operation, reflecting the light according to the pattern; a pellicle that is offset from the mask and is between the mask and the light source; and a pellicle assembly that, in operation, moves the pellicle with first motion that is independent of second motion of the mask.

[0092] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method, comprising: depositing a mask layer over a substrate; protecting a mask of a mask assembly by a pellicle attached to a pellicle assembly, the pellicle and the pellicle assembly being laterally offset from the mask assembly by a distance; directing first radiation toward the mask with the pellicle positioned overlapping the mask; and exposing the mask layer of a semiconductor wafer by second radiation carrying a pattern of the mask.

2. The method of claim 1, further comprising: moving the mask; and moving the pellicle by the pellicle assembly while the mask is moving, motion of the pellicle being independent of motion of the mask.

3. The method of claim 1, further comprising: accelerating the mask at a first rate; and accelerating the pellicle during the accelerating the mask, the accelerating the pellicle being at a second rate that is different than the first rate.

4. The method of claim 3, wherein the accelerating the pellicle at a second rate is accelerating the pellicle at the second rate that is less than the first rate.

5. The method of claim 1, further comprising: moving the mask; and during the moving the mask, holding the pellicle stationary.

6. The method of claim 1, further comprising: during the exposing the mask layer, moving the pellicle by an actuator of the pellicle assembly.

7. The method of claim 1, further comprising: forming a first opening in the mask layer by removing a pattern region of the mask layer based on the pattern; forming a second opening by removing at least a portion of a layer underlying the mask layer exposed by the first opening; and forming a feature of a semiconductor device in the second opening.

8. A method, comprising: exposing a first semiconductor wafer by performing a first exposure operation using a first mask and a first pellicle, the first pellicle being attached to a pellicle assembly, the first pellicle and the pellicle assembly being laterally offset from the first mask by a distance, the pellicle assembly including an actuator; removing the first pellicle from the pellicle assembly; after the removing, attaching a second pellicle to the pellicle assembly, the second pellicle being a different pellicle than the first pellicle; and exposing a second semiconductor wafer by performing a second exposure operation using the second pellicle.

9. The method of claim 8, wherein the exposing a second semiconductor wafer includes the performing a second exposure operation using a second mask different than the first mask.

10. The method of claim 8, further comprising: prior to the removing the first pellicle, transferring the pellicle assembly and the first pellicle to a lock and load chamber.

11. The method of claim 10, further comprising: cleaning the first pellicle by venting the lock and load chamber.

12. The method of claim 10, further comprising: increasing a pressure in the lock and load chamber to a first level about atmospheric pressure.

13. The method of claim 12, further comprising: after the increasing the pressure, transferring the pellicle assembly from the lock and load chamber to a first chamber.

14. The method of claim 10, further comprising: after the attaching a second pellicle, moving the pellicle assembly from the first chamber to the lock and load chamber.

15. The method of claim 14, further comprising: after the moving the pellicle assembly, lowering the pressure of the lock and load chamber to a second level that is at vacuum or near-vacuum; and after the lowering pressure, moving the pellicle assembly to an exposure position in which the second pellicle is between a light source and the first mask.

16. A system, comprising: a light source that, in operation, generates light; a mask having a pattern, the mask, in operation, reflecting the light according to the pattern; a pellicle that is offset from the mask and is between the mask and the light source; and a pellicle assembly that, in operation, moves the pellicle with first motion that is independent of second motion of the mask.

17. The system of claim 16, wherein the pellicle assembly includes an actuator.

18. The system of claim 16, wherein the pellicle is offset from the mask by a distance in a range of about 1 millimeter to about 3 millimeters.

19. The system of claim 16, further comprising: a second chamber that, in operation, is at a vacuum or near-vacuum pressure, the mask being in the second chamber during exposing a semiconductor wafer by the light reflected therefrom; a first chamber that, in operation, is at a pressure higher than that of the second chamber; and a lock and load chamber that, in operation: is at the vacuum or near-vacuum pressure while the lock and load chamber receives the pellicle assembly and the pellicle from the second chamber; and is at the pressure while the lock and load chamber transfers the pellicle assembly and the pellicle to the first chamber.

20. The system of claim 16, wherein the pellicle assembly is further operable to hold the pellicle stationary while the mask is moving.